

1 Data Structure of Orders

This document describes each of the fields for an order header. The actual sequence of the editing dialogs and reports/overviews may deviate from the one illustrated here.

In order to simplify matters, the term order will generally be used, regardless of whether an order or a work plan is being discussed. Only in examples in which it would make sense for the overall understanding to differentiate between the two will we use the term work plan.

General tab

Order / work plan

The order number or rather the work plan number is an upper-level number, under which each of the operations is compiled.

Order type

Order types are issued to structure the orders in accordance with their use. Each order type includes various control information that is decisive when managing orders.

The glossary describes the order types that are available by default in the system. You can define additional order types when customizing the system.

Article/Item

Material number/ item number/ article numbers of the (final) article to be produced with this order. If no article is entered for the operation, the system transfers the article included in this field to the operations.

Article name

Name of the article. Any changes to the article name are transferred to all operations of the order (redundant information). You cannot edit the article name in relation to operations.

Drawing issue number

Drawing issue number of the article, also referred to as index (available as of BDE 8.2).

Customer name

Customer name

Sales order

Sales order number

Sales order item In addition to the sales order number, you can also enter the line item number of the sales order that this order refers to.

Priority

You can use the "priority" as a control tool for the order. The priority is a single digit, numeric value. The value increases in ascending order ("0" = lowest priority, "9" = highest priority).

This value specifies the color for the operation bar in the graphic planning board (Shop Floor Scheduling). The colored bar graphically indicates the priority of the operation. You can assign the colors to the priorities in the graphic planning board settings of the Shop Floor Scheduling product group.

During the customization process, you can determine based on the order type

- whether the priority of the order header is transferred unchanged to each of the operations
- whether priority management should be enabled.

Order index

The order index can be seen as an alternative to the priority. You can integrate the order index, for example, in sorting the graphic detailed planning (if configured accordingly).

The *order index* is numerical with a valid value range (-999.9 to +999.9).

Target quantity

Quantity specification for the production order in base quantity unit. The indicated target quantity may include a target scrap quantity that might have been entered.

Target scrap

Planned scrap quantity for the production order in base quantity unit. The indicated scrap quantity can be considered as part of the transferred target quantity.

Unit

Quantity unit of the order for the (final) article to be produced. The unit allows you to compare, for example, scrap from different operations. That is why, the unit is included as a base quantity unit in each operation (redundant information).

Material type

Material type of the (final) article to be produced. If the field does not include a material type, the MES inserts the value "SYSTEM" here.

Batch number

The batch number reserved for the order; is generally provided by the ERP system.

Dates tab

Basic start date

The basic start date of the order. In general, the ERP system specifies this date.

Basic end date

The basic end date of the order. In general, the ERP system specifies this date. This date is based on the required date/ delivery date set in the ERP system and, if necessary, also includes buffer times.

Scheduled start time

Scheduled start date; result of the lead-time scheduling as compared to infinite capacities.

If the scheduling is run outside of the system, the scheduled dates in the order header should be applied. If the scheduling is run in MES, these fields are overwritten.

Scheduled end time

Scheduled end date; result of the lead-time scheduling as compared to infinite capacities.

If the scheduling is run outside of MES, the scheduled dates in the order header should be applied. If the scheduling is run in MES, these fields are overwritten.

Scheduling type

The scheduling type describes whether the order is scheduled forward (V) or backward (R) during lead-time scheduling in MES. If scheduled forward, the order is scheduled based on the basic start date specified in the ERP system. If scheduled backward, the order is scheduled based on the specified basic end date.

If no scheduling type is set in the order, the scheduling type defined in *Basic Settings* is used.

Reduction strategy

If it turns out during scheduling that the lead time for a given order is longer than the allotted time available, then MES will attempt to take reduction measures to shorten the lead time accordingly. Reducible times are the waiting times and the transport times.

The document hls-bk.doc describes in the chapter *Reduction Strategies* how to configure reduction strategies. The configuration is performed as part of the customization process.

Assignment tab**Order group**

If the field *order group* includes a value and the [Priority control](#) is enabled, the system checks the following when you attempt to create a new order:

- how many orders do exist for this order group and the specified priority in the system
- does this new order exceed the limit defined in the MOC application [Order groups](#). If this new order exceeds the specified limit, the system will reject the order. If the priority control function is enabled, you must:
 - consider the order group as a mandatory field in the order and
 - define a preset value range in the MOC application *order groups*.

Note for SAP users

The term *order group* corresponds to the SAP production scheduler. To ensure a consistent data exchange between SAP and HYDRA, you should synchronize the possible SAP production schedulers with the MES order groups.

MRP controller

The MRP controller for the order. You can transfer the MRP controller from SAP to the MES for informational purposes. You can display the MRP controller in the MES. But the MES does not provide a predefined value range.

Project number

Project order number

Planned order

Planned order number, e.g. in serial production.

Cost object

Cost object number

Work plan

Work plan number of the work plan that served as the template for generating the production order.

Work plan version

Version number of the work plan that served as the template for generating the production order.

BOM version

Version of the bill of material assigned to the production order.

Production version

Production version on which the order is based. This field is currently only completed if planned orders are transferred via the HKMPP-REM interface.

Closed loop

ID of the closed loop / supply relationship for which a Kanban order has been generated.

Inspection order

Inspection order/ inspection batch number for the order

Sample type

Type of sample for the order

Calculation tab

The *calculation* index tab includes additional data fields where calculation-related values or information can be stored. These entries are for information purposes only.

Machine costs

Calculated value for the machine costs that are incurred in the production of this order.

Labor costs

Calculated value for the labor costs that are incurred in the production of this order.

Material costs

Calculated value for the material costs that are incurred in the production of this order.

Other costs

Calculated value for other costs that are incurred in the production of this order.

Material value

Calculated value of the produced final article for each base quantity unit.

Scrap value

Calculated scrap value for each base quantity unit.

User fields tab

User fields allow you to store further customer-specific information to the MES in addition to the fields that are available by default. The order information shows the order-related user fields. The order information dialog provides the *user fields* index tab for the order header. This tab shows the user field key, the defined user fields including the names and units of measure. The user fields tab includes eight sub-index tabs, which each have eight additional user fields. The so-called user field key determines which user fields are involved and which meaning they have.

User field key

Every user field key describes a combination of user fields. The management of the user field key (and therefore the purpose of the fields) varies from one object to the next.

User fields

The following user fields are available for the order header (object type AUNR) after customizing the system:

Field ID / index	Field data type	Number of fields
1 - 6	Date	6
7 - 22	Numeric, time, duration	16
23 - 28	Decimal value	6
29 - 44	Text field, length 1	16
45 - 50	Text field, length 10	6
51 - 64	Text field, length 20	14
65 - 66	Text field, length 40	2

User field keys are defined in coordination with the customer during the customization process.

Administration tab

The administration index tab includes technical information on the data record. The dialogs "Insert" and "Copy" do not provide this index tab.

Created by

User who created the order.

Created on

Time and date when the order was created.

Modified by

User who most recently changed the order header.

Modified on

Time and date when this modification was made.

Transferred by

Here, you can enter the source from where the order was transferred.

Transferred on /Transfer time

If the ERP system transfers the order (PPS=J), the system automatically sets the transfer time and date to the time and date when the order was stored in MES.

Modified HYDRA

Specifies that the order was modified in MES. This identifier is automatically set to "J", if the order was changed in MES.

Modified PPS

Specifies that the production order was changed in the ERP system. This identifier is automatically set to "J", if the production order was changed in the ERP system (PPS=J). The identifier is not reset.

Deletion flag

Used for internal processing purposes. Cannot be modified.

Responsibility area

If a responsibility area is entered here, the user must have been authorized to view and edit orders and/or work plan orders.